Department of Physics
National Institute of Technology Calicut

Mid-Semester Examination

PH7302 – Condensed Matter Physics (2026–27)

Roll No & Name: _____

Duration: 2 Hours **Maximum Marks: 45**

General Instructions.

1. Section A: 5 MCQs (Q1–Q5, 1 mark each; all compulsory).
2. Section B: Attempt **any three** of Q6–Q9 (8 marks each, $3 \times 8 = 24$ marks).
3. Section C: Q10 is **compulsory**; answer **any one** from Q11 or Q12 (8 marks each, $2 \times 8 = 16$ marks).
4. No negative marking in Section A.
5. Physical constants: $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$, $\hbar = 1.054 \times 10^{-34}\,\mathrm{J\,s}$, $e = 1.602 \times 10^{-19}\,\mathrm{C}$, $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$, $k_\mathrm{B} = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$, $N_\mathrm{A} = 6.022 \times 10^{23}\,\mathrm{mol^{-1}}$, $1\,\mathrm{eV} = 1.602 \times 10^{-19}\,\mathrm{J}$, $1\,\mathrm{amu} = 1.66 \times 10^{-27}\,\mathrm{kg}$.

Section A — Multiple Choice Questions

[Questions 1–5. $5 \times 1 = 5$ marks. All questions are compulsory.]

1. A plane has intercepts $4a$, $2b$, ∞c. Its Miller indices are:

   (a) $(4, 2, 0)$ (b) $(\frac{1}{4}, \frac{1}{2}, 0)$ (c) $(1, 2, 0)$ (d) $(0, 1, 2)$

2. The packing fraction of an FCC crystal is:

   (a) 0.52 (b) 0.68 (c) 0.74 (d) 0.91

3. In a monoatomic chain, the group velocity at the Brillouin-zone boundary is:

   (a) $\sqrt{C/M}\,a$ (b) $\sqrt{C/M}\,a/2$ (c) 0 (d) $\sqrt{4C/M}$

4. The Dulong–Petit high-T molar heat capacity is:

   (a) $3R$ (b) R (c) $3Nk_\mathrm{B}T$ (d) $3Nk_\mathrm{B}/T$

5. The Drude conductivity can be recovered from the quantum picture because:

   (a) the equilibrium part dominates
   (b) only electrons near E_F contribute, moving at v_F
   (c) all electrons drift at v_th
   (d) the Fermi surface vanishes at low T

Section B

*[Answer any **three** from Questions 6–9. $3 \times 8 = 24$ marks.]*

6. **[2+3+3]**

(a) Define the reciprocal lattice vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ through the condition $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$ and write their explicit cross-product expressions.

(b) Show that the reciprocal lattice vector $\mathbf{G}_{hkl}$ is normal to the (hkl) family of lattice planes, and derive the relation $d_{hkl} = 2\pi/|\mathbf{G}_{hkl}|$.

(c) Copper has an FCC structure with lattice parameter $a = 3.61\,\text{Å}$. Compute the interplanar spacing d_{220}. Using Bragg's law with X-ray wavelength $\lambda = 1.54\,\text{Å}$, calculate the glancing angle θ for first-order (220) reflection.

7. **[2+3+3]**

(a) Define the structure factor F_{hkl}, explicitly writing its mathematical expression $F_{hkl} = \sum(expression)$ and defining all terms. Explain how F_{hkl} determines whether a reflection (hkl) is allowed or systematically absent based on $I_{hkl} \propto |F_{hkl}|^2$.

(b) For a BCC lattice with basis $(0,0,0)$ and $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$, derive the expression $F_{hkl} = f[1 + e^{\pi i(h+k+l)}]$ and state the systematic extinction rule.

(c) For an FCC lattice with basis $(0,0,0)$, $(\frac{1}{2},\frac{1}{2},0)$, $(\frac{1}{2},0,\frac{1}{2})$, and $(0,\frac{1}{2},\frac{1}{2})$, derive the extinction rule. Specify which of the following reflections are observed: (100), (110), (111), (200), (210), and (220).

8. **[4+1+3]**

(a) Derive the dispersion relation $\omega(k) = 2\sqrt{C/M}\,|\sin(ka/2)|$ for a 1-D monoatomic lattice of identical atoms of mass M separated by distance a, assuming harmonic nearest-neighbor interactions with spring constant C.

(b) Identify the expressions for the speed of sound v_s and the cutoff frequency ω_max.

(c) For a 1-D chain with $M = 40\,\text{amu}$, $a = 3\,\text{Å}$, and $C = 50\,\text{N m}^{-1}$, compute ω_max, v_s, and the group velocity v_g at $k = 0.3\pi/a$.

9. **[3+2+3]**

(a) Set up the equations of motion for a 1-D diatomic chain of masses M and m connected by identical springs with force constant C. Formulate the secular determinant and quote the resulting dispersion relation.

(b) Show that at the edge of the first Brillouin zone, an energy gap $\Delta\omega = \sqrt{2C/m} - \sqrt{2C/M}$ opens up between the acoustic and optical branches.

(c) Taking $M = 40\,\text{amu}$, $m = 20\,\text{amu}$, and $C = 50\,\text{N m}^{-1}$, compute the gap width $\Delta\omega$, the acoustic edge frequency $\omega_-(\pi/a)$, and the optical edge frequency $\omega_+(\pi/a)$.

Section C

*[Question 10 is compulsory. Answer any **one** from Q11 or Q12. $2 \times 8 = 16$ marks.]*

10. **[4+1+3]**

(a) Derive the 3-D Debye density of vibrational states $g(\omega) = 9N\omega^2/\omega_{\mathrm{D}}^3$, where the Debye cutoff frequency is defined as $\omega_{\mathrm{D}} = v_{\mathrm{s}}(6\pi^2 N/V)^{1/3}$.

(b) Starting from the internal energy, show that the low-temperature lattice heat capacity follows the T^3 law: $C_V = \frac{12\pi^4}{5} N k_{\mathrm{B}} (T/\Theta_{\mathrm{D}})^3$.

(c) Sodium has a Debye frequency $\nu_{\mathrm{D}} = 3.3 \times 10^{12}\,\mathrm{Hz}$. Compute its Debye temperature $\Theta_{\mathrm{D}} = h\nu_{\mathrm{D}}/k_{\mathrm{B}}$ and its molar specific heat at $T = 10\,\mathrm{K}$.

11. **[4+2+2]**

(a) Using periodic boundary conditions for a free electron gas in 3 dimensions, derive the density of states $D(E) = \frac{V}{2\pi^2}(\frac{2m}{\hbar^2})^{3/2}\sqrt{E}$ and the Fermi energy $E_{\mathrm{F}} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$.

(b) Sodium (monovalent, BCC) has density $\rho = 970\,\mathrm{kg\,m^{-3}}$ and atomic weight $A = 23\,\mathrm{g\,mol^{-1}}$. Compute the conduction electron density n, the Fermi energy E_{F} in eV, and the Fermi temperature T_{F}.

(c) Calculate the average kinetic energy per electron at $T = 0\,\mathrm{K}$ and estimate the fraction of conduction electrons excited within an energy range $\pm k_{\mathrm{B}}T$ of E_{F} at $T = 300\,\mathrm{K}$.

12. **[4+2+2]**

(a) Using the steady-state Boltzmann transport equation within the relaxation-time approximation, derive the expression for the electrical conductivity $\sigma = ne^2\tau/m$.

(b) Copper has a conduction electron density $n = 8.5 \times 10^{28}\,\mathrm{m^{-3}}$ and relaxation time $\tau = 2.5 \times 10^{-14}\,\mathrm{s}$. Compute its electrical conductivity σ and the drift velocity v_{d} in an applied electric field $E = 1\,\mathrm{V\,m^{-1}}$.

(c) Compute the Fermi velocity $v_{\mathrm{F}} = \hbar(3\pi^2 n)^{1/3}/m$ and the electronic mean free path $\ell = v_{\mathrm{F}}\tau$. Compare this mean free path with the interatomic spacing ($a \approx 3.61\,\text{Å}$).

Question-wise CO Mapping and Mark Distribution

Question Nos.	1	2	3	4	5	6	7	8	9	10	11	12
Course Outcomes	CO1	CO1	CO2	CO2	CO3	CO1	CO1	CO2	CO2	CO2	CO3	CO3
Difficulty Level*	1	1	2	1	2	3	3	3	4	3	3	3
Marks	1	1	1	1	1	8	8	8	8	8	8	8

Course Outcomes:
CO1: Differentiate between different lattice types and explain the concepts of reciprocal lattice and crystal diffraction.
CO2: Distinguish different excitations in crystals and explain the specific heat.
CO3: Apply the free electron model and interpret the properties of metals.
CO4: Explain the concept of energy bands and use it to find electrical properties of materials (*End-sem examination*).
CO5: Explain electrical and magnetic properties of materials (*End-sem examination*).

Difficulty Level*:
1 = Knowledge / Recall level;
2 = Understand / Comprehend level;
3 = Apply / Analyse level;
4 = Evaluate / Create level.

Notes:
(i) Section A (Q1–Q5): Compulsory ($5 \times 1 = 5$ marks).
(ii) Section B (Q6–Q9): Attempt *any three* ($3 \times 8 = 24$ marks).
(iii) Section C: Q10 is compulsory (8 marks); attempt *any one* from Q11 or Q12 (8 marks) ($2 \times 8 = 16$ marks).
(iv) **Total Marks = 45** (Section A: 5 + Section B: 24 + Section C: 16).

End of Question Paper. Best of luck!